

Homework 1

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Problem 1.

(a) **A1: DFS on fixed puzzle**

- Start state: 283164705
- Goal state: 123804765
- Number of moves: 5
- Path found:
283164705 $\rightarrow$ 283104765 $\rightarrow$ 203184765 $\rightarrow$ 023184765 $\rightarrow$ 123084765
 $\rightarrow$ 123804765

(b) **B1: A* on fixed puzzle**

- Start state: 283164705
- Goal state: 123804765
- Number of moves: 5
- Path found (same as A1 in this case):
283164705 $\rightarrow$ 283104765 $\rightarrow$ 203184765 $\rightarrow$ 023184765 $\rightarrow$ 123084765
 $\rightarrow$ 123804765

(c) **C1: Parity test**

- Original start parity: 1
- After swapping first two numbers (2 and 8): parity becomes 0

- Goal parity: 1
- Conclusion: swapped start state and goal state have different parity, so they are unreachable.

Problem 2.

(a) **A2/B2 random states (seed = 225040065)**

- Random start state: 718532046
- Random goal state: 584732106
- The two random states are not reachable directly (parity mismatch), so I swapped a neighboring pair in the start state.
- Adjusted start state: 178532046

(b) **A2: DFS on random puzzle (after adjustment)**

- Start state: 178532046
- Goal state: 584732106
- Number of moves: 57545
- Simplified path (key states):
178532046 → 178032546 → 078132546 → ... → 584032716 →
584732016 → 584732106

(c) **B2: A* on random puzzle (after adjustment)**

- Start state: 178532046
- Goal state: 584732106
- Number of moves: 21
- Simplified path (key states):
178532046 → 178532406 → 178502436 → ... → 584730162 →
584732160 → 584732106